

MCA and Gower distances

Categorical data · Mixed types ·
Completing the toolkit

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Today's Agenda

Completing the EDA toolkit: categorical and mixed-type data

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Mixed data
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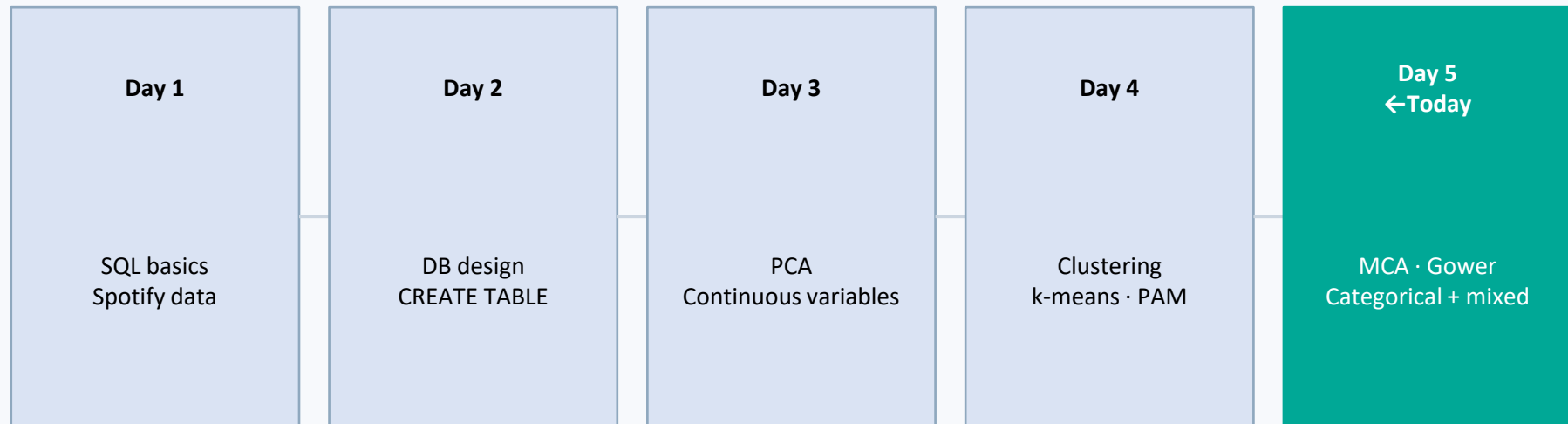
05

Synthesis
+ Lab

Decision framework
All five methods
ESS lab

Where We Are: The Final Piece

Days 3–4 covered continuous data; today we handle when data is really mixed up



The dataset today is the European Social Survey: ordinal and nominal survey responses from >2000 respondents across 22 countries. The same data is used for both MCA and Gower, showing that the two methods complement rather than replace each other.

Today's data: European Social Survey (ESS)

A cross-national academic survey used throughout this lecture and lab

What is the ESS?

The **European Social Survey** is a rigorous cross-national academic survey conducted every two years since 2002. It covers countries across Europe using probability sampling and face-to-face interviews.

Here, we use a selection of individual-level responses from Round 10 (2020), which covers ~37,000 respondents across 26 countries.

Our lab dataset is a stratified sample of 100 respondents per country (22 countries) = 2,175 respondents in total (with some NA:s removed).

Freely available at <https://www.europeansocialsurvey.org/> after registration

Our variable selection

Political trust (active)

trust_parliament · trust_legal · trust_police
trust_politicians · trust_parties

Social attitudes (active)

pro_immigration · religiosity

Satisfaction (active)

life_satisfaction · satis_economy · satis_government

Supplementary variables

cuntry · left_right · gender · age_group · education

All 10 active variables were originally 0–10 numeric scales, but here discretised to low / medium / high (0–3 → low, 4–6 → medium, 7–10 → high) before analysis.

1

From PCA to MCA

Why categorical data needs a different approach



Why PCA Cannot Handle Categorical Variables Directly

The geometric assumption breaks down!

PCA finds directions of maximum variance and computes means. Both operations require variables to be continuous and interval-scale. Categorical variables violate this: the mean of {low, medium, high} doesn't make sense, and the variance of {male, female} has no geometric interpretation.

Nominal variables have no order

Country codes (BE=1, BG=2, CH=3...) look numeric but the numbers are arbitrary labels. Computing a mean or variance of country codes is meaningless. PCA would treat BE as closer to BG than to FR purely because of the alphabetical coding.

Ordinal variables have no unit

low/medium/high can be coded 1/2/3, but the gap between low and medium may not equal the gap between medium and high. PCA assumes equal intervals, which is a coding choice that drives the result. (Then again, MCA treats all ordinals as nominals)

Frequencies, not magnitudes

With categorical variables the interesting question is not 'what is the average trust level?' but 'which categories tend to co-occur in the same respondents?' That is an association question, not a variance question.

MCA solves this by converting categorical variables into a binary matrix first, then applying a form of PCA to that matrix. Same geometric intuition, but with different mathematical machinery under the hood! (Though issues with ordinal variables...)

The MCA Idea: so-called Complete Disjunctive Table

Convert each categorical variable into a set of binary columns with their possible values, then apply PCA-like approach!

Original data

Resp.	trust	politics
R1	low	left
R2	high	right
R3	medium	centre
R4	low	right



Complete Disjunctive Table (CDT)

Resp.	trust_low	trust_med	trust_high	pol_left	pol_centre	pol_right
R1	1	0	0	1	0	0
R2	0	0	1	0	0	1
R3	0	1	0	0	1	0
R4	1	0	0	0	0	1

Each categorical variable with k categories becomes k binary columns. Each row has exactly one 1 per original variable. A variable with 3 categories contributes 3 columns. With 11 variables and 3 categories each: 33 binary columns total. THIS MEANS THAT VALUE ORDER ACTUALLY DOES NOT MATTER FOR MCA! Any ordinal factor variable is actually treated as nominal under the hood!

PCA-style approach is applied to the CDT

The geometric logic is identical to PCA: find axes of maximum variance in the binary matrix. The math is slightly different (chi-square distances rather than Euclidean) but the output has the same structure.

Categories become coordinates

Each category (e.g. trust_low, trust_high) gets a position in the factor space. Categories that tend to appear together in the same respondents end up close together on the map.

Inertia instead of variance

MCA uses 'inertia' rather than variance. The total inertia is determined by the number of variables and categories. Dim1 explains the most inertia, Dim2 the next most. Same idea as in PCA.

MCA Output: Same Geometry, Different Data

Category coordinates instead of variable loadings

	PCA	MCA
Input	Observations × continuous variables	Observations × categorical variables (via CDT)
Key output 1	Individual scores (where <u>observations</u> sit)	Individual scores (where <u>respondents</u> sit)
Key output 2	Variable loadings (correlation circle with variable vectors)	Category coordinates (each category level is a point on the map)
Supplements	quanti.sup, quali.sup, ind.sup	Same! quanti.sup, quali.sup, ind.sup
Explained variance	% variance per component: often 40–70% for PC1+PC2	% inertia per dimension: often 15–30% for Dim1+Dim2 (expected to be lower)
Interpretation	Name each axis using variable loadings	Name each axis using which categories cluster together

The low % inertia in MCA is not a sign of failure: it is expected and does not mean the solution is poor. With many binary columns, variance is distributed across more dimensions by construction. (Check out the so-called ‘Benzecri correction’ if you need comparable numbers)

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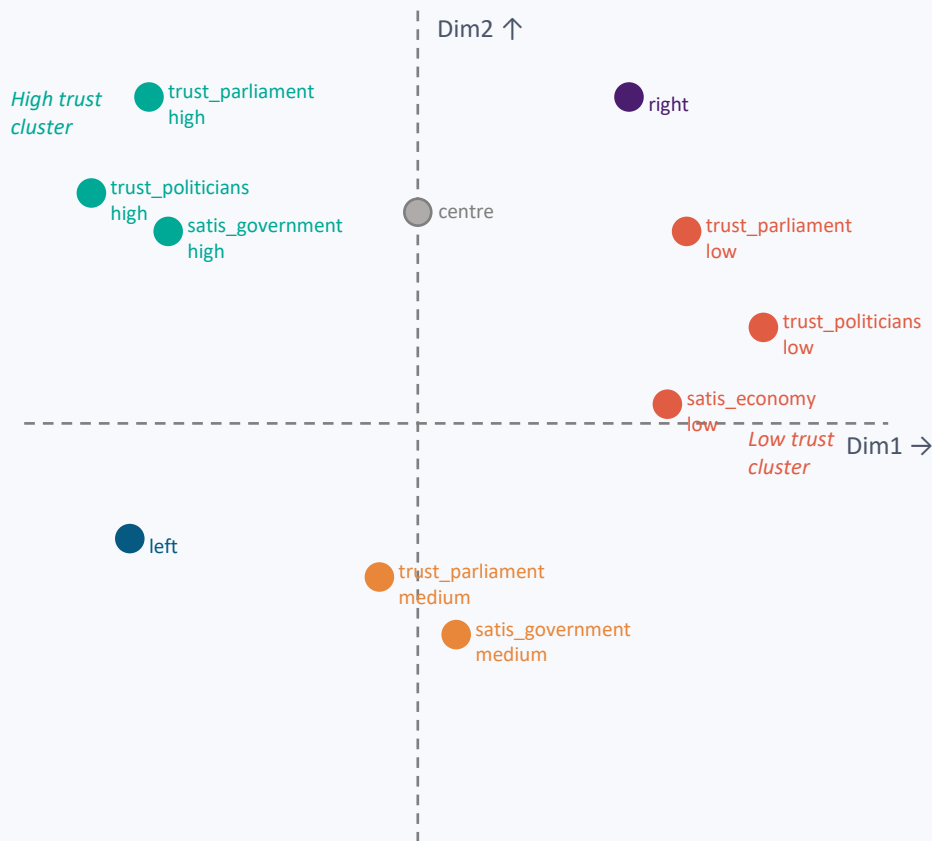
Reading MCA output

Category map · Correlation plot · Supplementaries



The Category Coordinate Map

Each category level is a point; proximity means co-occurrence. Note: combination variable AND value!



How to read the category map:

Proximity = co-occurrence

Categories that are close together tend to appear in the same respondents. `trust_parliament_high` near `satis_government_high` means high trusters tend to also be satisfied with government.

Distance from origin

Points far from the origin are rare or distinctive — they pull the dimensions. Points near the origin are common but not discriminating.

Opposite positions = opposition

`trust_high` on one side and `trust_low` on the other: these categories are carried by different respondents and define what the axis means.

"Medium" near centre

Medium categories tend to sit near the origin in MCA.

Name the axes

Look at which categories are at the extremes of Dim1 and Dim2. High trust + high satisfaction on one end, low trust + low satisfaction on the other → Dim1 is an institutional confidence axis.

The Correlation Plot and Supplementary Variables

Which variables drive which dimension (and projecting external information)

```
fviz_mca_var(mca_res, choice = 'mca.cor')
```

Shows the squared correlation between each variable and each dimension (not the individual category positions but the variable as a whole)

All points land in the positive quadrant (correlations are squared → always 0–1). Variables far from the origin on the x-axis drive Dim1. Variables far up the y-axis drive Dim2.

Use this plot to name the axes: the variables farthest from the origin along each axis are the ones that give the dimension its meaning.

This is the MCA equivalent of the PCA contribution plot.

Supplementary variables in MCA

quali.sup: categorical supplements

Country, gender, age group. Projected as category points without influencing the axis computation. Seeing where Norway, Bulgaria, and Sweden land on the trust map is a key result, and it is more convincing because they did not shape the axes.

quanti.sup: continuous supplements

Age in years, years of education. Projected as vectors like in PCA. Shows how continuous variables correlate with the MCA dimensions.

left_right in the ESS data

Left-right self-placement as a supplementary variable is pedagogically elegant: does the MCA solution correspond to the conventional left-right axis? If it aligns with Dim1 or Dim2, that dimension has an ideological interpretation.

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MCA in R

MCA() · fviz_mca_ · ESS data workflow*



MCA in R: Complete Workflow

From loading the ESS data to reading the factor map

```
library(FactoMineR); library(factoextra); library(tidyverse)

# Read the csv file (no SQL this time)
ess <- read_csv("data/ess10_lab.csv")

# Then we need to fix the 'factors'! These are ordinals, not just strings. Check the R script for how this can
# be done

# We drop the 'idno' unique identifiers: nothing to visualize or include in MCA
ess <- ess |> select(-idno)

# Then we do the MCA! Note how similar it is to PCA.
# We also indicate the specific columns that we DON'T want to include
mca_res <- MCA(ess_mca,
  quali.sup = c(1, 12, 13, 14, 15),
  graph      = FALSE)

# Once done, we can see how much inertia explained per dimension (similar to explained variance in PCA)
summary(mca_res)
```

See: `lab_mca_ess.R`

Visualising MCA Results

```
# 1. Which variables drive each dimension?
fviz_mca_var(mca_res, choice = "mca.cor", repel = TRUE)
# Variables far right → drive Dim1; variables far up → drive Dim2

# 2. Where do category levels sit? (main interpretive plot)
fviz_mca_var(mca_res, repel = TRUE,
  col.var      = "red",
  col.quali.sup = "darkgreen")
# Red = active category coordinates (trust_parliament_low, _medium, _high etc.)
# Green = supplementary categories (countries, age groups, gender)
# Proximity = co-occurrence in same respondents
```

See: `lab_mca_ess.R`

Visualising MCA Results

```
# 3. Individuals coloured by a supplementary variable
fviz_mca_ind(mca_res,
             geom.ind = "point",
             col.ind = ess_mca$age_group,
             palette = "jco",
             alpha.ind = 0.4,
             addEllipses = TRUE,
             legend.title = "Age group")

# Try other than ess_mca$age_group! See lab script

# 4. Biplot – individuals + category coordinates together
fviz_mca_biplot(mca_res,
               geom.ind = "point",
               alpha.ind = 0.15,
               repel = TRUE)

# With 2100ish respondents: use alpha.ind to reduce overplotting
```

See: `lab_mca_ess.R`



Break

4

Gower distance

Mixed data · How it works · PAM + Gower in R



The Mixed Data Problem

Real social science data is almost never one clean type

Most datasets you will work with in computational social science contain a mix of variable types in the same rows. Euclidean distance requires continuous data. MCA requires categorical data. What do you do when you have both?

A typical dataset — one row per country, multiple variable types:

Country	GDP per capita (SEK)	Life exp.	Region	Income group	OECD
nominal	continuous	continuous	nominal	ordinal	binary
Norway	82,000	83.2	N.Europe	high	yes (1)
Bulgaria	11,500	75.1	E.Europe	upper-middle	no (0)
Brazil	9,800	76.4	S.America	upper-middle	no (0)
Nigeria	2,200	62.1	W.Africa	lower-middle	no (0)

PCA + Euclidean

→ uses: GDP per capita, life expectancy

Ignores region, income group, OECD membership: loses half the information

MCA

→ uses: Region, income group (nominal), OECD

Ignores continuous indicators: also loses half the information

Gower distance

→ uses: All variables together

Handles each type appropriately: nothing dropped/lost!

How Gower Distance Works

For each pair of observations, Gower computes a dissimilarity score for each variable separately using a formula appropriate for that variable type, then averages across all variables. The result is always between 0 and 1.

Variable type	Dissimilarity formula	Example (Norway vs Bulgaria)	Score
Continuous	$ x_A - x_B / \text{range}(x)$	GDP: $ 82000 - 11500 / 80000 = 0.88$	0.88
Ordinal	$ \text{rank}_A - \text{rank}_B / (n_{\text{levels}} - 1)$	Income: $ \text{high} - \text{upper-mid} / 3 = 0.33$	0.33
Nominal	0 if same category, 1 if different	Region: N.Europe $\neq$ E.Europe $\rightarrow 1.0$	1.00
Binary	0 if both 1 or both 0, 1 if different	OECD: yes $\neq$ no $\rightarrow 1.0$	1.00

$$\text{Gower}(\text{Norway, Bulgaria}) = (0.88 + 0.33 + 1.00 + 1.00) / 4 = 0.80$$

Average across 4 variables (each gets equal weight)

Handles missing values gracefully

If a variable is NA for either observation, simply exclude it from the average for that pair (example above, divide by 3). No imputation needed!

Always between 0 and 1

Each variable contribution is normalised before averaging. No variable can dominate because it has large numbers (unlike Euclidean distance)

PAM + Gower in R: Complete Workflow

Using the ESS data you already know

```
library(cluster)

# Load and prepare data (as for MCA: see script)

# Step 1: Prepare, now including demographics as active variables (as it is possible)
ess_gower <- ess_mca |>
  select(cntry, trust_parliament:left_right,
         gender, age_group, education) |>
  drop_na()

# Step 2: Compute Gower distance
# daisy() detects variable types automatically from factor/numeric classes
gower_dist <- daisy(ess_gower, metric = "gower")
```

See: `lab_gower_ess.R`

diss = TRUE is essential when passing a pre-computed distance matrix to `pam()`. Without it PAM tries to treat the matrix as raw data and produces nonsense. The flip side: k-means cannot accept a distance matrix at all (which is why Gower + PAM is the natural pairing)

PAM + Gower in R: Complete Workflow

Using the ESS data you already know

```
# Step 3: Choose k - calc silhouette and plot it (look for an elbow)
sil_widths <- map_dbl(2:8, function(k) {
  pam_k <- pam(gower_sample, k = k, diss = TRUE)
  mean(silhouette(pam_k)[, "sil_width"])
})

fviz_nbclust(ess_gower, FUN = function(x, k) pam(daisy(x,"gower"), k),
             method = "silhouette", k.max = 8)

# Step 4: PAM - diss=TRUE because input is a distance matrix
pam_ess <- pam(gower_dist, k = 4, diss = TRUE)
ess_gower[pam_ess$id.med, ] # medoid respondents
```

See: `lab_gower_ess.R`

diss = TRUE is essential when passing a pre-computed distance matrix to `pam()`. Without it PAM tries to treat the matrix as raw data and produces nonsense. The flip side: k-means cannot accept a distance matrix at all — which is why Gower + PAM is the natural pairing.

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Synthesis and Lab

Decision framework · All five days · ESS exercises



A Method Decision Framework

Given your normal or crazy data: which method?

Your data	Dimensionality reduction	Clustering	Example in this course
All continuous variables (same scale)	PCA scale.unit = FALSE	k-means or hierarchical + Euclidean	<i>Valkalkylator party responses</i>
All continuous variables (different scales)	PCA scale.unit = TRUE	k-means or hierarchical + Euclidean	<i>Spotify genre profiles</i>
All categorical / ordinal variables	MCA	PAM + Gower or k-modes	<i>ESS survey attitudes</i>
Mixed: continuous + categorical + ordinal	PCA + MCA separately, or skip	PAM + Gower (handles all types)	<i>Check out World Bank country indicators</i>

1

The method follows from the data

You don't choose PCA because you like it. You choose it because your data meets its assumptions! Choosing MCA for continuous data or PCA for nominal categories is methodologically wrong.

2

You can always use Gower!

Gower with PAM is always valid regardless of data type. For all-continuous data it gives similar results to Euclidean + PAM. Its flexibility makes it the safe default when you are unsure. (But PCA would be better if applicable!)

3

Combine where possible

MCA first (for structure in the categorical space) then PAM + Gower (to group respondents formally) is a natural two-step workflow. The MCA dimensions help interpret the Gower clusters.

Lab 5: MCA and Gower on ESS Data

Two methods, same dataset, complementary insights

Part A

MCA ESS Attitudes

- Load and prepare `ess10_lab.csv`
- Set factor levels in correct order
- Run `MCA()` with `quali.sup`
- Plot `mca.cor`: which variables drive each dim?
- Plot `var.cat`: name Dim1 and Dim2
- Colour individuals by age group or country

`lab_mca_ess.R`

Part B

Gower + PAM ESS Respondents

- Include demographics as active variables
- Compute Gower distance with `daisy()`
- Choose `k` with silhouette plot
- Run `pam(..., diss = TRUE)`
- Inspect medoid respondents — who are they?
- Compare cluster profiles across trust variables

`lab_gower_ess.R`

Part C

Compare MCA vs Gower

- Add MCA Dim1 score to clustered data frame
- Do Gower clusters correspond to MCA regions?
- Which method gives more interpretable groups?
- Write: one paragraph describing what divides respondents
- What does each method add that the other misses?

`lab_gower_ess.R`

Thank you!

